Exam Equations Physics II

Trigonometry: $\sin\Theta = h_o/h$ $\quad \cos\Theta = h_a/h$ $\quad \tan\Theta = h_o/h_a$

$\Theta = \sin^{-1}(h_o/h)$ $\quad \Theta = \cos^{-1}(h_a/h)$ $\quad \Theta = \tan^{-1}(h_o/h_a)$

Pythagorean Theorem $h^2 = h_o^2 + h_a^2$

$\cos(90° - \Theta) = \sin(\Theta)$ $\quad \cos(90° + \Theta) = -\sin(\Theta)$ $\quad \cos(180° - \Theta) = -\cos(\Theta)$

$\sin(90° - \Theta) = \cos(\Theta)$ $\quad \sin(90° + \Theta) = \cos(\Theta)$ $\quad \sin(180° - \Theta) = \sin(\Theta)$

Vectors: $\vec{\mathbf{a}} = (a_x)\hat{\mathbf{x}} + (a_y)\hat{\mathbf{y}}$ $\quad \vec{\mathbf{a}} \bullet \hat{\boldsymbol{x}} = a_x = |\vec{a}|\cos(\alpha)$ $\quad \vec{\mathbf{a}} \bullet \hat{\boldsymbol{y}} = a_y = |\vec{a}|\cos(\beta)$

$$\alpha = \cos^{-1}\left[\frac{a_x}{|\vec{\mathbf{a}}|}\right] \qquad \beta = \cos^{-1}\left[\frac{a_y}{|\vec{\mathbf{a}}|}\right] \qquad |\vec{\mathbf{a}}| = \sqrt{\vec{\mathbf{a}} \bullet \vec{\mathbf{a}}} = \sqrt{(a_x)^2 + (a_y)^2}$$

Dot product: $\vec{a} \bullet \vec{b} = |\vec{\mathbf{a}}||\vec{\mathbf{b}}|\cos(\varphi)$

Displacement: $\vec{\boldsymbol{x}} = \vec{r}_f - \vec{r}_i$ $\quad$ Speed = d/t $\quad$ velocity: $\vec{\mathbf{v}} = d\vec{\mathbf{x}}/dt$ $\quad$ acceleration: $\vec{a} = d\vec{\mathbf{v}}/dt$

Newton's Laws of Motion $\Sigma\vec{\mathbf{F}} = m\vec{\boldsymbol{a}}$ or $\vec{\boldsymbol{a}} = \Sigma\vec{\boldsymbol{F}}/m$ $\quad$ Weight: $\vec{W} = m\vec{\boldsymbol{g}}$, $\quad |\vec{\boldsymbol{g}}| = 9.8$ N/kg

$\Sigma\vec{\mathbf{F}} = \vec{\mathbf{F}}_{\mathbf{net}} = (\Sigma F_x)\hat{\boldsymbol{x}} + (\Sigma F_y)\hat{\boldsymbol{y}}$ where $\Sigma F_x = ma_x$ and $\Sigma F_y = ma_y$

For equilibrium: $\Sigma F_x = 0$ and $\Sigma F_y = 0$

For non-equilibrium: $\Sigma F_x = ma_x$ and $\Sigma F_y = ma_y$

Kinetic Energy: $KE = \frac{1}{2}mv^2$ $\quad$ Gravitational Potential Energy: $PE_g = mg(y_f - y_i) = mgh$

Elastic Potential Energy: $PE_e = \frac{1}{2}kx^2$, $\quad$ Springs: $F = k\Delta x$,, $\quad$ Conservation of Energy, $\Delta KE + \Delta PE = 0$

Coulomb's law: $\vec{\mathbf{F}}_E = (kq_1q_2/r^2)\hat{\mathbf{r}}$ (in vacuum), $k = 9.0 \times 10^9\ Nm^2/C^2$ $\quad$ Charge: $Q = Nq_e$,

Uniform Electric Field: $|\vec{E}| = |\vec{\mathbf{F}}_E|/q$, $\quad F_E = qE$, $\quad$ Electric Field due to point charge: $\vec{\mathbf{E}} = (kq/r^2)\,\hat{\mathbf{r}}$

$$Charge\ distribution \quad \vec{E} = \left(k\int \frac{1}{r^2}dq\right)\hat{r} \qquad \vec{E} = (E_x)\hat{x} + (E_y)\hat{y} + (E_z)\hat{z}$$

Linear charge density: $\lambda = Q/L$ $\quad dq = \lambda dL$ $\quad$ Surface charge density: $\sigma = Q/A$ $\quad dq = \sigma dA$

Volume charge density: $\rho = Q/V$ $\quad dq = \rho dV$

Total charge: $Q = \int dq$

Electric Flux: Uniform $\vec{E}$ $\quad \Phi_E = \vec{E} \bullet \vec{A}$ $\qquad$ Area vector $= \vec{A} = (|\vec{A}|)\hat{n}$ $\quad$ Gauss's Law $\quad \Phi_E = \oint \vec{E} \bullet \hat{n}dA = \frac{Q_{in}}{\varepsilon_o}$

$k = 9.0 \times 10^9\ Nm^2/C^2$, $\quad \varepsilon_o = 8.85 \times 10^{-12}\ C^2/Nm^2$

Electron: $q_- = 1.6 \times 10^{-19}$ C, negative charge, $m_e = 9.1 \times 10^{-31}$ kg

Proton: $q_+ = 1.6 \times 10^{-19}$ C, positive charge, $m_p = 1.67 \times 10^{-27}$ kg

1m = 100 cm, 1 m = 1,000 mm, 1 cm = 10 mm, 1 micron (μ) = 10^{-6}, 1 nano (n) = 10^{-9}, 1 pico (p) = 10^{-12}

Sphere:

$$V_{Sphere} = \frac{4}{3}\pi R^3 \quad A_{Sphere} = 4\pi R^2$$

Cylinder:

$$V_{Cylinder} = \pi L R^2 \quad A_{Cylinder} = 2\pi RL + 2\pi R^2$$

Rectangular prism:

$$V_{Rectanglar} = Lwh \quad A_{Rectagular} = 2(Lw + Lh + wh)$$

Area of circle: $A_{circle} = \pi r^2$ Circumference of Circle: $C_{circle} = 2\pi r$ Area of a right triangle: $A_{Triangle} = \frac{1}{2}hb$
Angle Θ for circle: $\Theta = s/R$

<u>Rulers for Derivatives</u>

$$Power\ rule: \ \frac{d}{dx}[x^n] = nx^{n-1}$$

$$Addition\ or\ subtraction: \ \frac{d}{dx}[y \pm u] = \frac{d}{dx}[y] \pm \frac{d}{dx}[u] \quad Product\ rule: \ \frac{d}{dx}[yu] = u\frac{d}{dx}[y] + y\frac{d}{dx}[u]$$

$$\frac{\mathrm{d}}{\mathrm{dt}}[\sin(\omega t)] = \omega\cos(\omega t) \qquad \frac{\mathrm{d}}{\mathrm{dt}}[\cos(\omega t)] = -\omega\sin(\omega t) \qquad \frac{d}{dt}[e^{at}] = ae^{at}$$

<u>Integration Rules</u>

$$\int (0)\mathrm{dx} = \mathrm{C} \qquad \int (1)\mathrm{dx} = \mathrm{x} + \mathrm{C} \qquad \int (\mathrm{mx})\mathrm{dx} = \mathrm{m}\int (x)dx \qquad where\ C, a\ and\ m\ are\ constants$$

$$Addition\ and\ subtraction: \ \int (u+v)dx = \int (u)dx + \int (v)dx$$

$$Power\ Rule \quad \int x^n\,dx = \frac{x^{n+1}}{n+1} \qquad \int \sin(mx)\,dx = -\frac{\cos(mx)}{m} \qquad \int \cos(mx)\,dx = \frac{\sin(mx)}{m} \qquad \int e^{at}dt = \frac{1}{a}e^{at}$$

$$Definate\ Integral: \ \int_a^b [Integrand]dx = [Integral]\,|_a^b = \text{Integral with (b)} - \text{Intergral with (a)}$$

$$\int \frac{1}{x}dx = ln[x] \qquad \int \frac{1}{(x^2 \pm a^2)^{\frac{1}{2}}}dx = ln\left[x + \sqrt{x^2 \pm a^2}\right] \qquad \int \frac{x}{(x^2 \pm a^2)^{\frac{1}{2}}}dx = \sqrt{x^2 \pm a^2}$$

$$\int \frac{1}{(x^2 + a^2)^{3/2}}dx = \frac{x}{a^2(x^2 + a^2)^{1/2}} \qquad \int \frac{x}{(x^2 + a^2)^{3/2}}dx = -\frac{1}{(x^2 + a^2)^{1/2}}$$